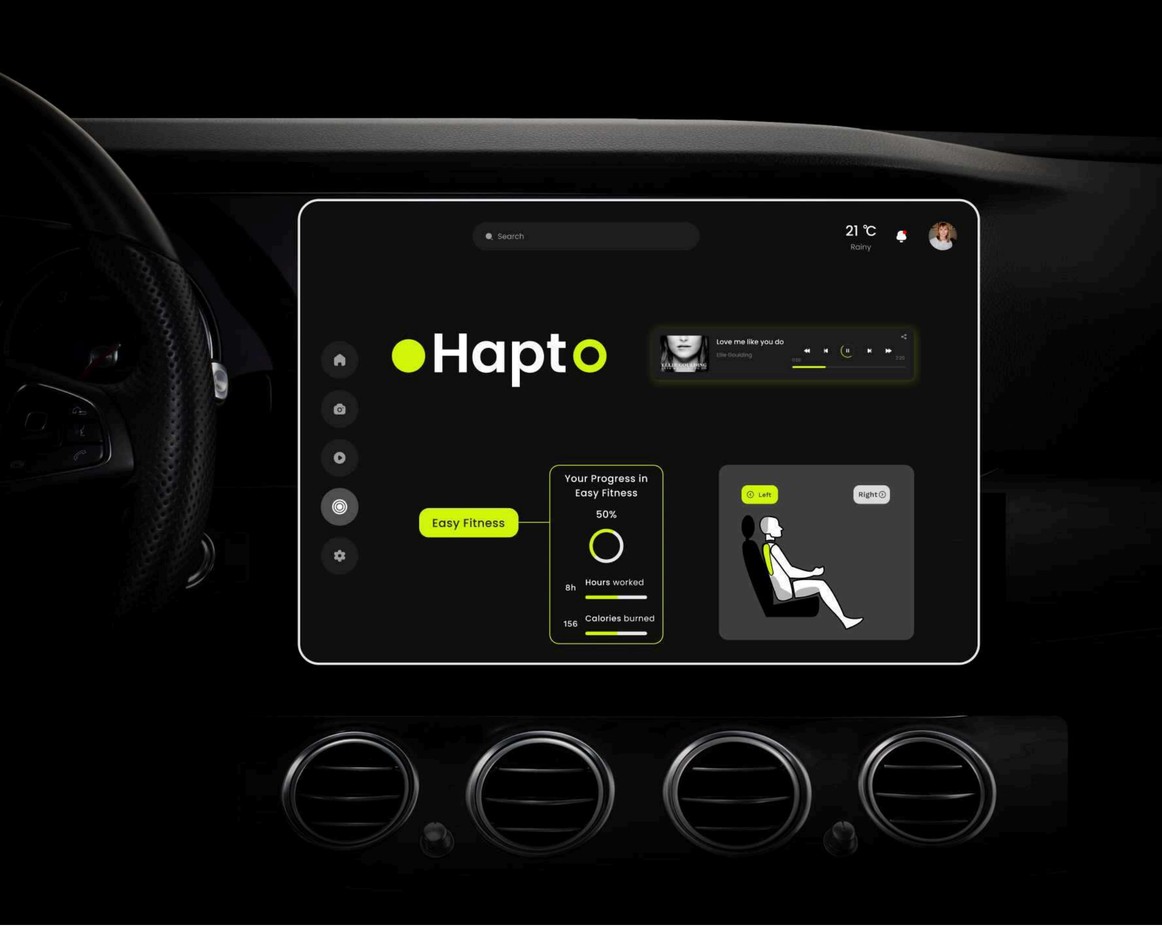

Search
21 ℃
Rainy
Hapto
Love me like you do
Ellie Goulding
Easy Fitness
Your Progress in Easy Fitness
50%
8h
Hours worked
156
Calories burned
Left
Right

The project’s goal ↘

This project was developed **in collaboration with Innovobot, working with BBW** to explore the future of haptic feedback in automotive seating.
Future vehicles from BMW are expected to feature haptic car seats equipped with 10 embedded vibration actuators. These systems are designed to support safety, comfort, navigation, health, and entertainment through tactile feedback.

Challenge: Most drivers are unfamiliar with vibration parameters such as frequency, amplitude, and patterns, making it difficult for them to understand and personalize haptic feedback effectively.

Hapto
Navigation
You will get vibrating feedback support your navigation system.
BMW
Belt warning system
Switch On only in emergency case
Hapto
Innovobot

The project's goal ↘

Goal

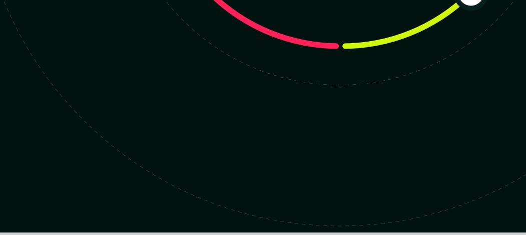
Design a testable, mid-fidelity prototype that enables users to intuitively control their haptic seat experience—without needing technical knowledge.

02
01
Hapto
03

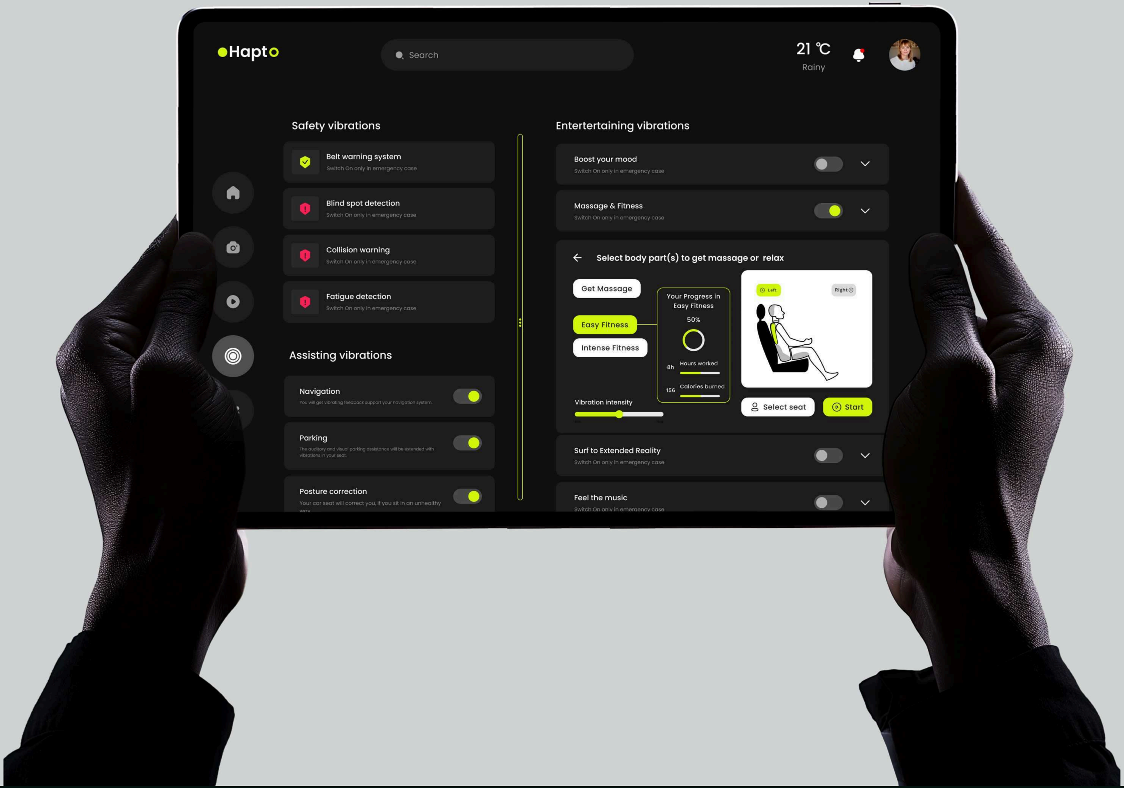
Question 1

What haptic aspects can users realistically understand and control?

Question 2



How can an interface present these controls in a user-friendly way?


Hapto
Search
21 ℃
Rainy
Get Massage
Easy Fitness
Intense Fitness
Your Progress in Easy Fitness
50%
8h
Hours worked
156
Calories burned
Vibration intensity
Left
Right
Select seat
Start

Safety vibrations

Belt warning system
Switch On only in emergency case

Blind spot detection
Switch On only in emergency case

Collision warning
Switch On only in emergency case

Fatigue detection
Switch On only in emergency case

Assisting vibrations

Navigation
You will get vibrating feedback support your navigation system.

Parking
The auditory and visual parking assistance will be extended with vibrations in your seat.

Posture correction
Your car seat will correct you, if you sit in an unhealthy way

Entertertaining vibrations

Boost your mood
Switch On only in emergency case

Massage & Fitness
Switch On only in emergency case

Select body part(s) to get massage or relax

Surf to Extended Reality
Switch On only in emergency case

Feel the music
Switch On only in emergency case

What did we do in this project? ↘

Phase 1

Exploratory Research

Examining haptic feadback in 5 domains, including Automotive, Aviation, Gaming chairs and Office seating

Phase 2

Focus on Automative

Identifyng five core haptic categories: Warning, Assistance, Massage & Health, Posture, and Entertainment.

Phase 3

Control parameters

Extracted four key haptic controllable parameters: Intensity, Location of vibration, Pattern, and Mode/Feature activation

Phase 4

Design

Identifying key UI elements and Design with purpose.

Phase 5

Test and Iteration

Conducting Usability Test and two times iteration

What did we do in phase 1? ↘

Finding showed that haptics are used for:
Safety notifications, Navigation cues, Comfort & health, Immersive entertainment

● Anti-vibration

cushions and
massaging features

Haptic

create an
immersive gaming
experience

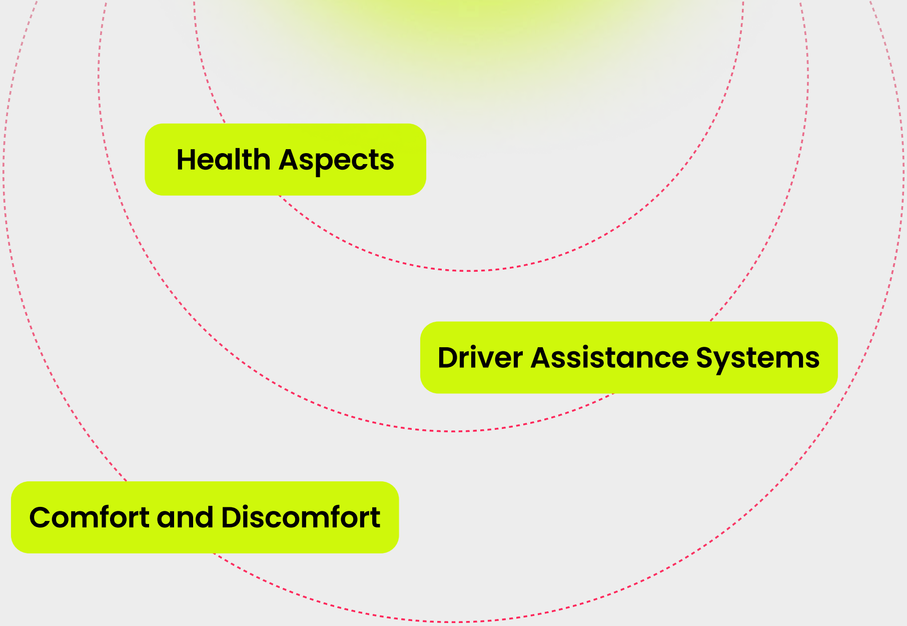
improve comfort
and ergonomics

media experiences,
Navigation, and improve
driver situational

awareness

What did we do in phase 1? ↘

We focused on automotive seats to explore how haptic feedback can enhance the seating experience, then concentrated on three key areas of its application in vehicle seating.



Health Aspects

Driver Assistance Systems

Comfort and Discomfort

Making User Stories ↘

We developed 10 user stories based on the so called three main areas. They served as the foundation for the UI design:

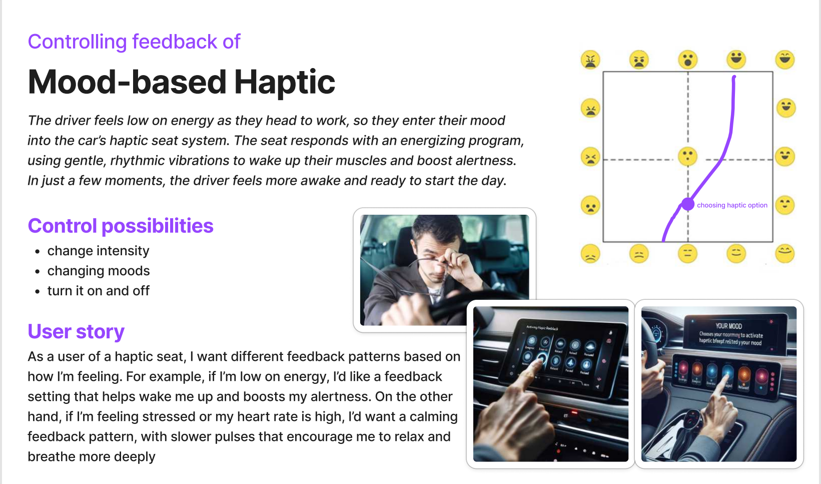
- Park Assistance
- Automatic Safety Assistance
- Navigation System
- Posture Correction Haptic
- Mood-Based Haptic
- Immersive Media Haptic

Forward Movement Assistance

In-Car XR

Mindful Driving

In-Car Fitness



Controlling feedback of

Mood-based Haptic

The driver feels low on energy as they head to work, so they enter their mood into the car's haptic seat system. The seat responds with an energizing program, using gentle, rhythmic vibrations to wake up their muscles and boost alertness. In just a few moments, the driver feels more awake and ready to start the day.

Control possibilities

- change intensity
- changing moods
- turn it on and off

User story

As a user of a haptic seat, I want different feedback patterns based on how I'm feeling. For example, if I'm low on energy, I'd like a feedback setting that helps wake me up and boosts my alertness. On the other hand, if I'm feeling stressed or my heart rate is high, I'd want a calming feedback pattern, with slower pulses that encourage me to relax and breathe more deeply

choosing haptic option
YOUR MOOD

Who can control the vibration? ↘

Government/Regulations

👤 **User/Driver**

🏢 **Car Manufacturer**

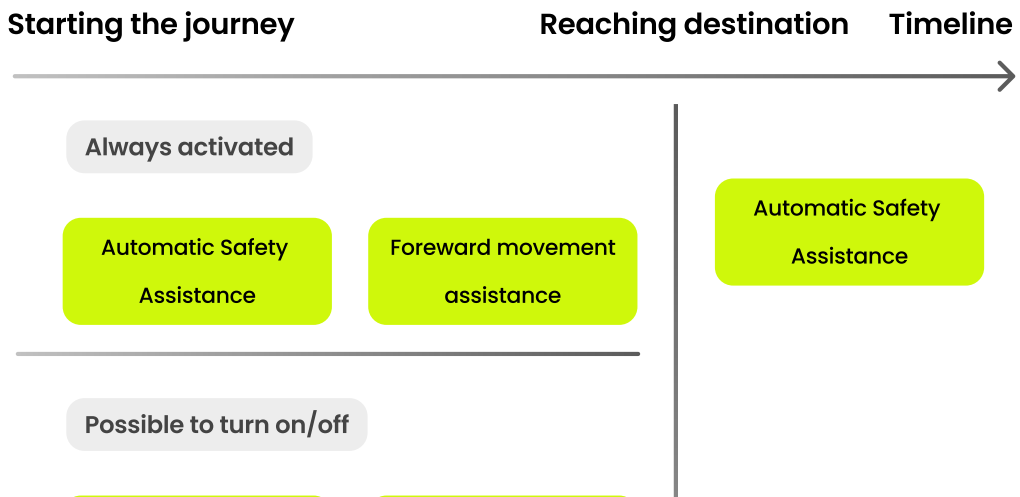
🤖 **Automation (AI)**

Controlling haptic parameters ↘

	Turn off/ on	Amplitude	Frequency	Location	Pattern
Forward movement assistant	⚖	⚖ 🤖	⚖	⚖	⚖
Park assistance	👤	👤	👤	🏢	🏢
Navigation system	👤	👤	👤	⚖	🏢
Automatic safety assistance	⚖	⚖ 🤖	⚖	⚖	⚖

Posture correction haptics	👤	🏢 🤖	🏢 🤖	👤 🤖	🏢
Mood-Based haptic	👤	👤	👤	🏢	🏢
In-car fitness	👤	👤 🤖	👤 🤖	👤 🏢	👤 🏢
Immersive media haptic	👤	👤	👤	🏢	🏢
In-car XR	👤	👤	👤	👤 🏢	👤 🏢

How is the prioritization? ↘



Starting the journey

Reaching destination

Timeline

Always activated

Automatic Safety Assistance

Foreward movement assistance

Automatic Safety Assistance

Possible to turn on/off

Navigation system

Posture correction haptics

One activated at a time

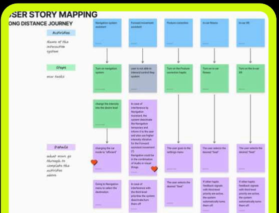
Mood-Based haptics

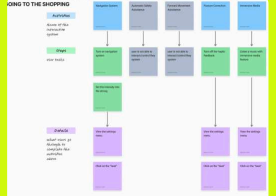
In-car fitness

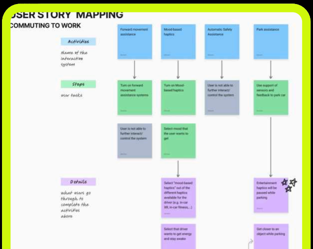
Immersive media haptics

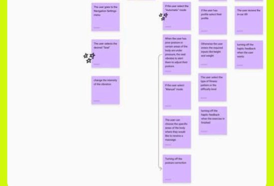
In-Car XR

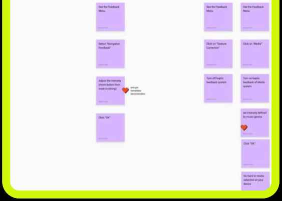
Making User Stories ↘

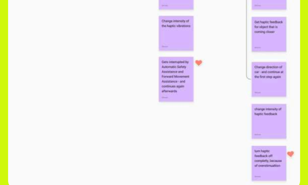

SER STORY MAPPING
ONG DISTANCE JOURNEY
Activities
Name of the interaction system
Steps
user tasks
Details
what users go through to complete the activities above


SER STORY MAPPING
OING TO THE SHOPPING
Activities
Name of the interaction system
Steps
user tasks
Details
what users go through to complete the activities above
Navigation System
Automatic Safety Assistance
Forward Movement Assistance
Posture Correction
Immersive Media
Turn on navigation system
Turn off the haptic feedback
View the settings menu
Click on the "Seat"

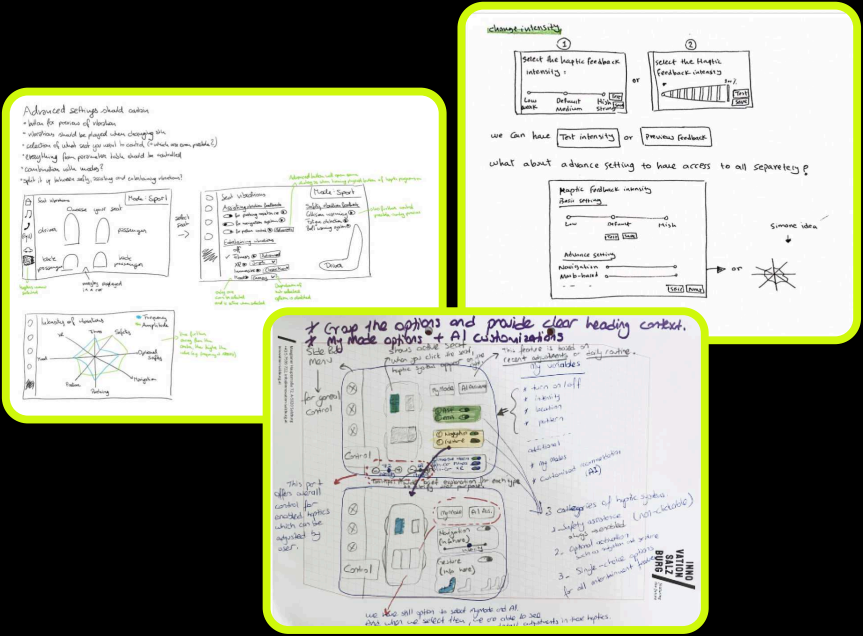

SER STORY MAPPING
COMMUTING TO WORK
Activities
Name of the interaction system
Steps
user tasks
Details
what users go through to complete the activities above
Forward movement assistance
Mood-based haptics
Automatic Safety Assistance
Park assistance
Turn on forward movement assistance systems
Turn on Mood-based haptics
User is not able to further interact/ control the system
Use support of sensors and feedback to park car
User is not able to further interact/ control the system
Select mood that the user wants to get
Select "mood-based haptics" out of the different haptics available for the driver (e.g. in-car XR, in-car fitness,...)
Entertainment haptics will be paused while parking
Select that driver wants to get energy and stay awake
Get closer to an object while parking





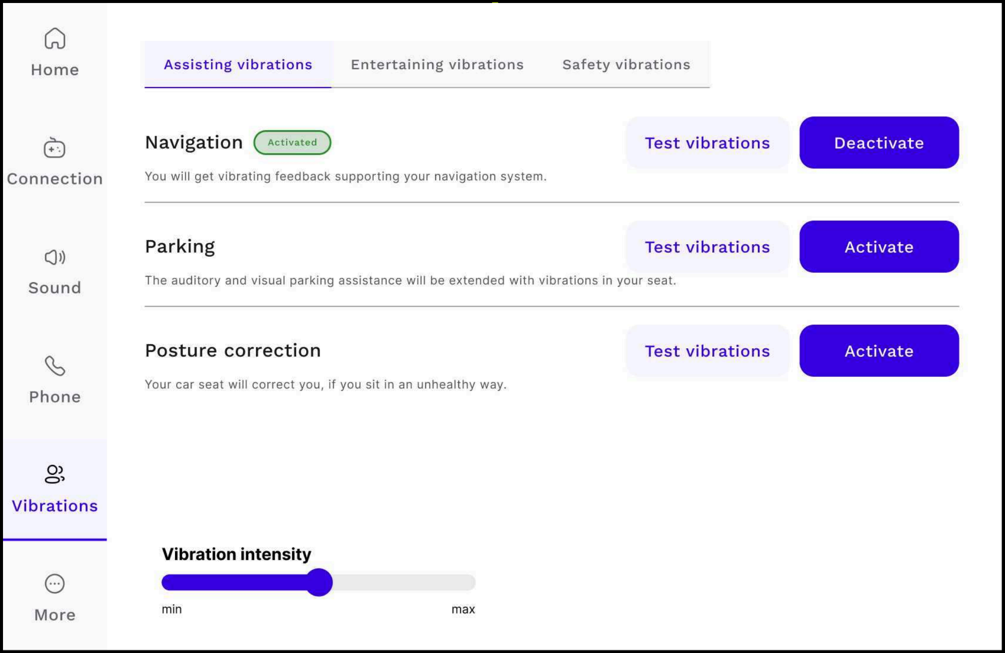


Early sketches ↘


change intensity
① Select the haptic feedback intensity:
Low weak
Default Medium
High Strong
Test
Save
or
② select the Haptic feedback intensity
100%
Test
Save
we can have Test intensity or Preview feedback
what about advance setting to have access to all separetely?
Haptic feedback intensity
Basic setting
Low
Default
High
Test
Save
Advance setting
Navigation
Mob-based
Skip
Next
or
Simone idea
Advanced settings should contain
• button for preview of vibration
• vibrations should be played when changing sth
• selection of what seat you want to control (→ which are even possible?)
• everything from parameter table should be controlled
• combination with modes?
• split it up between safety, assisting and entertaining vibrations?
Seat vibrations
Mode: Sport
Choose your seat
driver
passenger
back passenger
back passenger
maybe displayed in a car
select seat
Assisting vibration feedback
for parking assistance
for navigation system
for posture control
Advanced
Entertaining vibrations
off
Fitness
Advanced
XR
Jungle
Immersive
Classic Music
Mood
Energy
Safety vibration feedback
Collision warning
Fatigue detection
Belt warning system
Driver
Intensity of vibrations
Frequency
Amplitude
XR
Fitness
Safety
Optional Safety
Navigation
Parking
Posture
Mood
* Group the options and provide clear heading context.
* My mode options + AI customizations
Side Bar Menu
for general control
shows active seat, when you click the seat, the haptic system appear on the right
This feature is based on recent adjustments or daily routine.
My variables
* turn on/off
* intensity
* location
* pattern
additional
* my modes
* Customized recommendation (AI)
My Mode
AI Recovery
ASF
EMA
Navigation
Gesture
Control
This part offers overall control for enabled haptics which can be adjusted by user.
3 categories of haptic systems.
1- Safety assistance (non-clickable) always enabled
2- Optional activation such as navigation and gesture
3- Single-choice options for all entertainment features
My Mode
AI Ass.
Navigation (info here)
intensity
Gesture (info here)
Control
we have still option to select Mymode and AI.
And when we select them, we are able to see detail adjustments in these haptics.
INNO VATION SALZ BURG

Prototype ↘

Three main vibrations are showing on top menue. A key feature was a car visualization on the settings home screen, displaying all seats when activated. The ‘Massage&Fitness’ section introduced a body preview with three vibration modes (Figure 10), while ‘Boost Your Mood’, ‘Surf to Extended Reality’ , and ‘Feel the Beat’ featured multiple vibration modes with visualizations. Furthermore, for accessibility, the prototype included speech and gesture interactions, enabling voice and hand-gesture controls via a steering wheel button.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Test vibrations
Deactivate
Test vibrations
Activate
Test vibrations
Activate
min
max

Navigation Activated

You will get vibrating feedback supporting your navigation system.

Parking

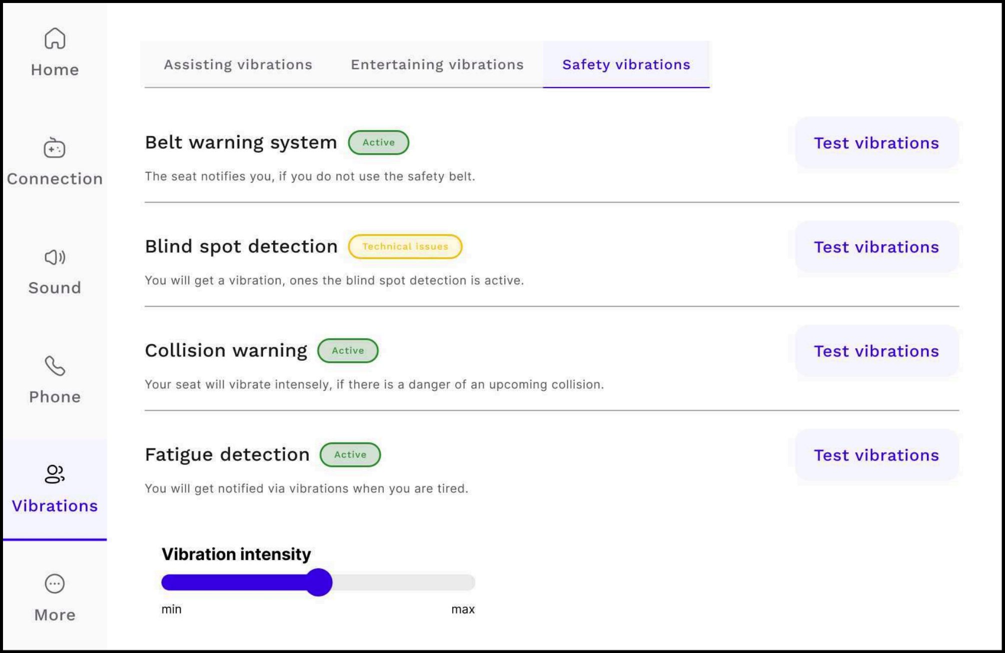
The auditory and visual parking assistance will be extended with vibrations in your seat.

Posture correction

Your car seat will correct you, if you sit in an unhealthy way.

Vibration intensity

It is possible to turn on/off Assisting Vibrations


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Test vibrations
Test vibrations
Test vibrations
Test vibrations
min
max

Belt warning system Active

The seat notifies you, if you do not use the safety belt.

Blind spot detection Technical issues

You will get a vibration, ones the blind spot detection is active.

Collision warning Active

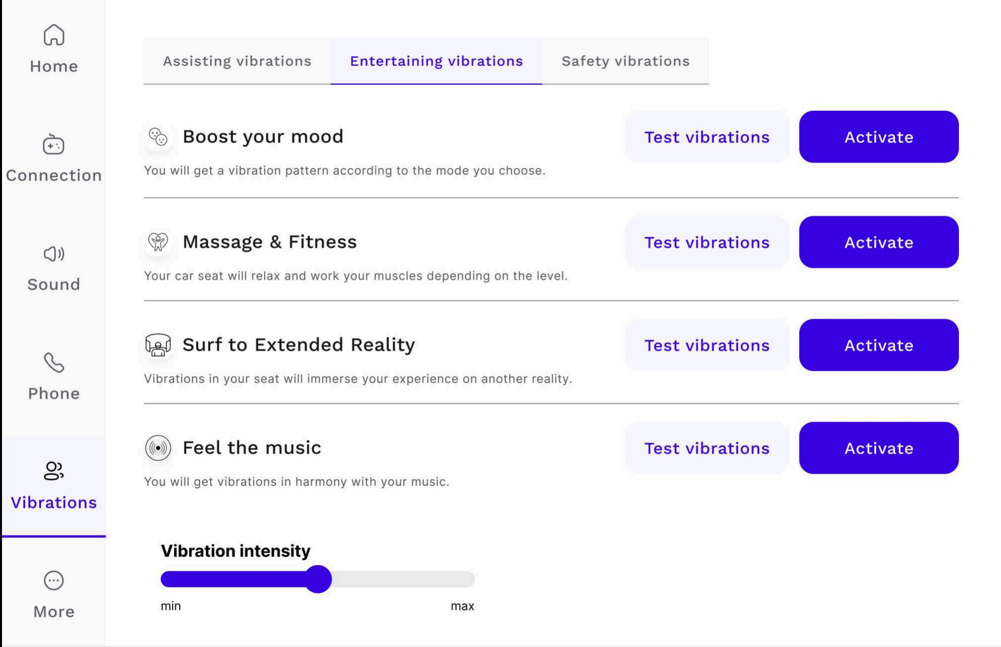
Your seat will vibrate intensely, if there is a danger of an upcoming collision.

Fatigue detection Active

You will get notified via vibrations when you are tired.

Vibration intensity

It is not possible to turn on/off Safety Vibrations


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Test vibrations
Activate
Test vibrations
Activate
Test vibrations
Activate
Test vibrations
Activate
Vibration intensity
min
max

Boost your mood

You will get a vibration pattern according to the mode you choose.

Massage & Fitness

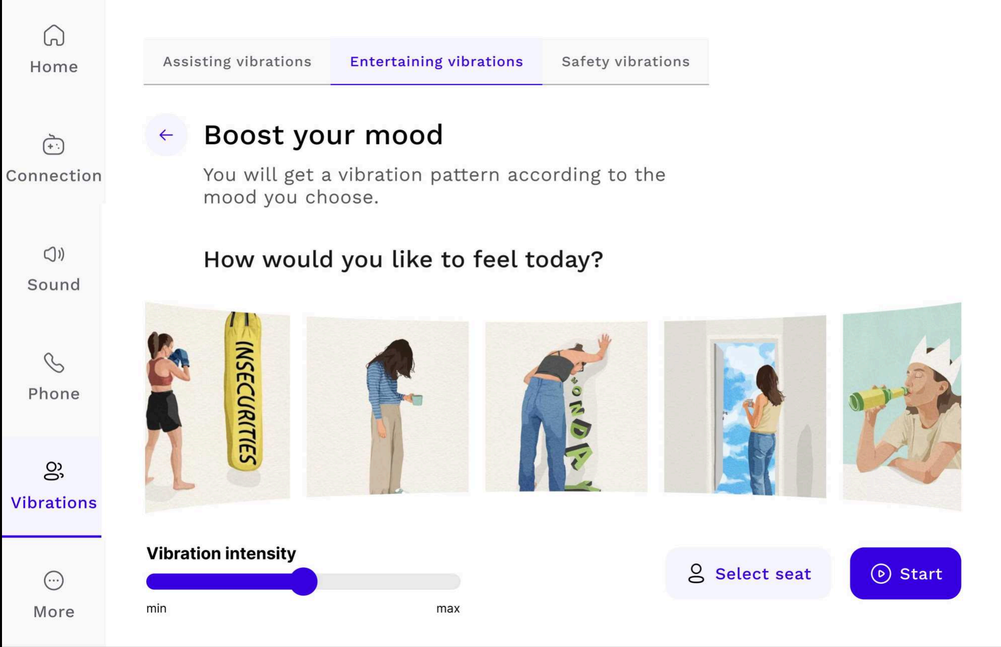
Your car seat will relax and work your muscles depending on the level.

Surf to Extended Reality

Vibrations in your seat will immerse your experience on another reality.

Feel the music

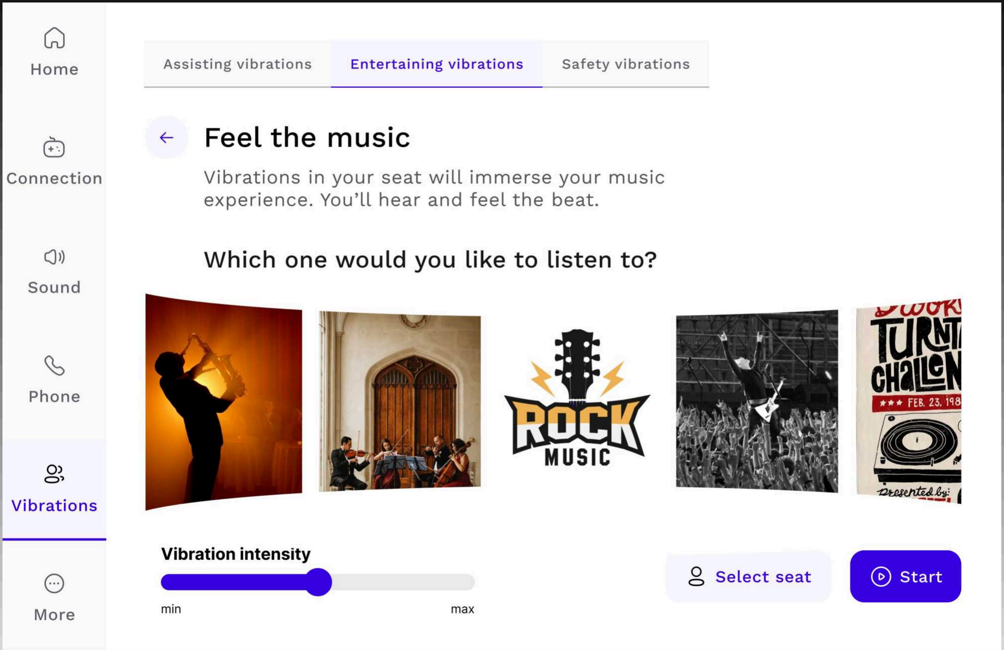
You will get vibrations in harmony with your music.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
←
INSECURITIES
MONDAY
Vibration intensity
min
max
Select seat
Start

Boost your mood

You will get a vibration pattern according to the mode you choose.

How would you like to feel today?

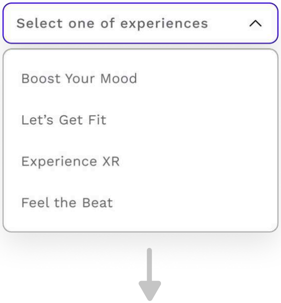

Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations

Feel the music

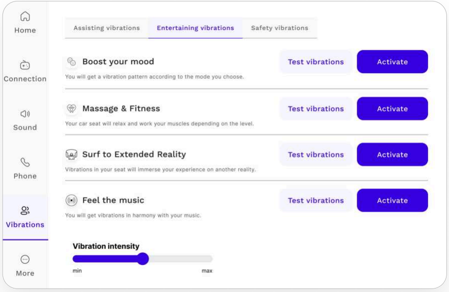
Vibrations in your seat will immerse your music experience. You'll hear and feel the beat.

Which one would you like to listen to?

ROCK MUSIC
TURNT CHALLEN
FEB. 23, 198
Presented by:
Vibration intensity
min
max
Select seat
Start


Select one of experiences
Boost Your Mood
Let's Get Fit
Experience XR
Feel the Beat

Replaced dropdown menus in the entertainment section with **tab-based navigation to improve accessibility** and make options easier for users to find and switch between.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Test vibrations
Activate
Test vibrations
Activate
Test vibrations
Activate
Test vibrations
Activate
min
max

Boost your mood

You will get a vibration pattern according to the mode you choose.

Massage & Fitness

Your car seat will relax and work your muscles depending on the level.

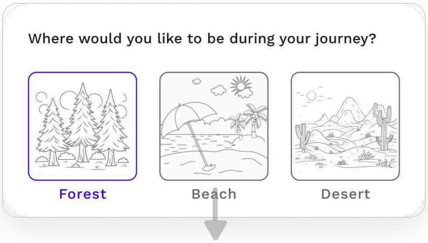
Surf to Extended Reality

Vibrations in your seat will immerse your experience on another reality.

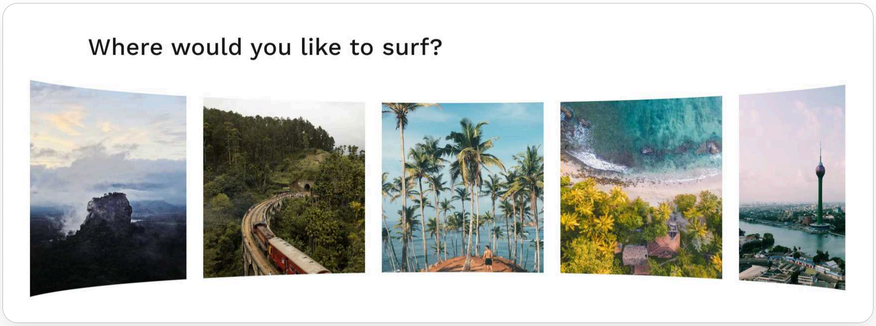
Feel the music

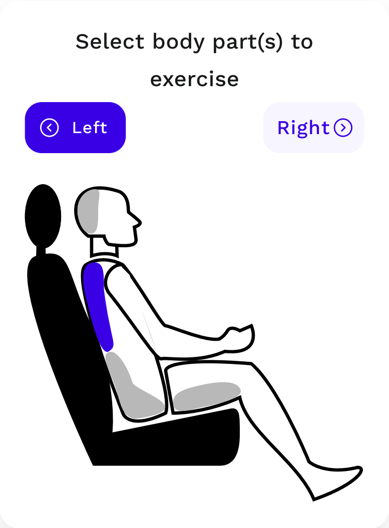
You will get vibrations in harmony with your music.

Vibration intensity


Where would you like to be during your journey?
Forest
Beach
Desert

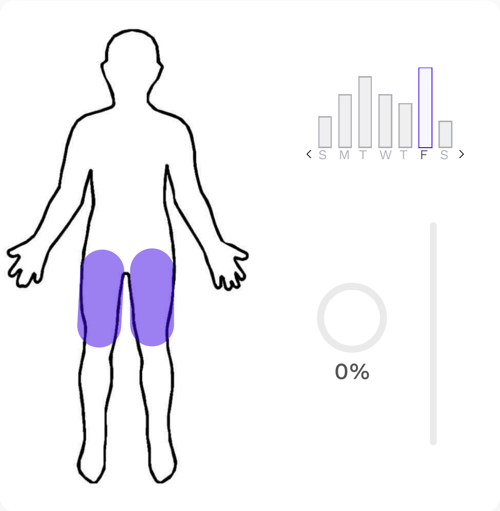
Using natural and **intuitive** pictures.


Where would you like to surf?



Select body part(s) to exercise

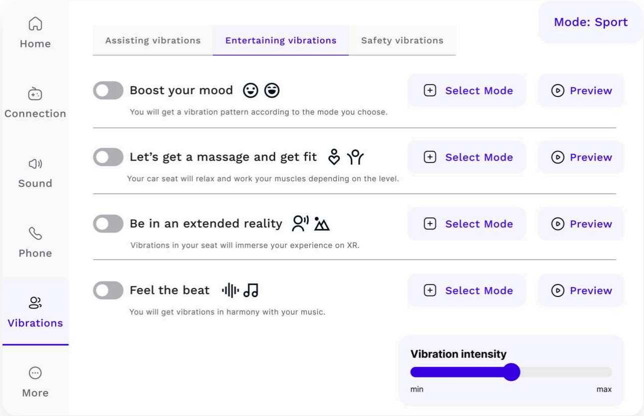
Display the **seated position** rather than standing, and highlight selectable body sections on the body map



Introduced **left and right buttons** so users can choose which side of the body they want to target.

Removed the toggle switch because it created confusion, especially since the safety vibrations were always active.

Replaced it with a clear Activate button.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Mode: Sport
Select Mode
Preview
Vibration intensity
min
max

Boost your mood

You will get a vibration pattern according to the mode you choose.

Let’s get a massage and get fit

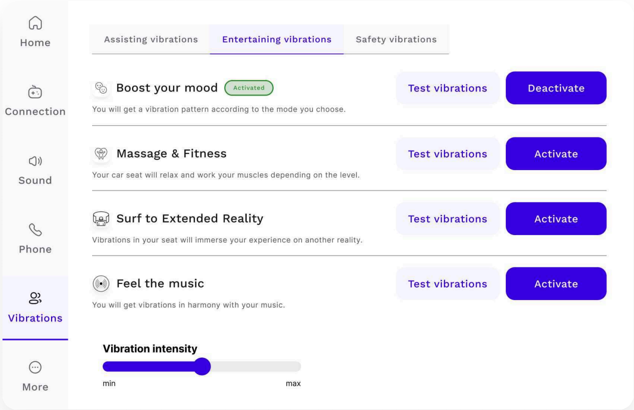
Your car seat will relax and work your muscles depending on the level.

Be in an extended reality

Vibrations in your seat will immerse your experience on XR.

Feel the beat

You will get vibrations in harmony with your music.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Activated
Test vibrations
Deactivate
Activate
Vibration intensity
min
max

Boost your mood

You will get a vibration pattern according to the mode you choose.

Massage & Fitness

Your car seat will relax and work your muscles depending on the level.

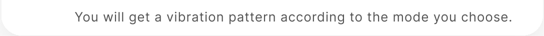
Surf to Extended Reality

Vibrations in your seat will immerse your experience on another reality.

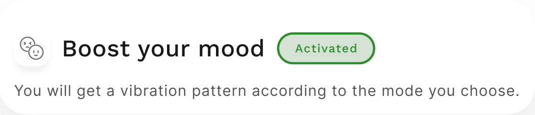
Feel the music

You will get vibrations in harmony with your music.

Boost your mood



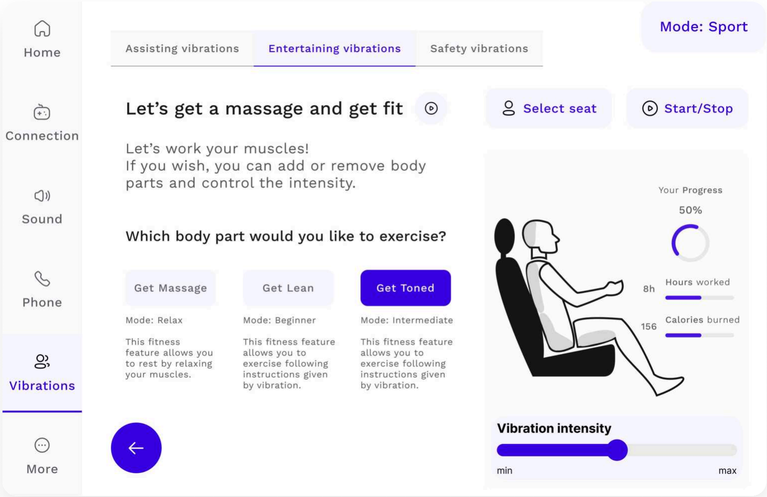
You will get a vibration pattern according to the mode you choose.



Boost your mood Activated

You will get a vibration pattern according to the mode you choose.

Added descriptive labels next to section headings to improve clarity, **including an Activated status label** to clearly indicate when vibration is active.



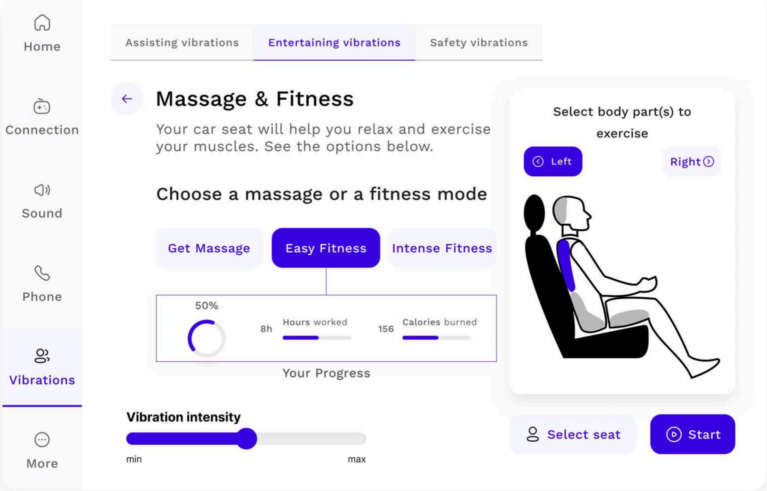
Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Mode: Sport
Select seat
Start/Stop
Your Progress
50%
8h
Hours worked
156
Calories burned
Vibration intensity
min
max

Let's get a massage and get fit

Let's work your muscles!
If you wish, you can add or remove body parts and control the intensity.

Which body part would you like to exercise?

Get Massage	Get Lean	Get Toned
Mode: Relax	Mode: Beginner	Mode: Intermediate
This fitness feature allows you to rest by relaxing your muscles.	This fitness feature allows you to exercise following instructions given by vibration.	This fitness feature allows you to exercise following instructions given by vibration.


Home
Connection
Sound
Phone
Vibrations
More
Assisting vibrations
Entertaining vibrations
Safety vibrations
Get Massage
Easy Fitness
Intense Fitness
50%
8h
Hours worked
156
Calories burned
Your Progress
Vibration intensity
min
max
Left
Right
Select seat
Start

Massage & Fitness

Your car seat will help you relax and exercise your muscles. See the options below.

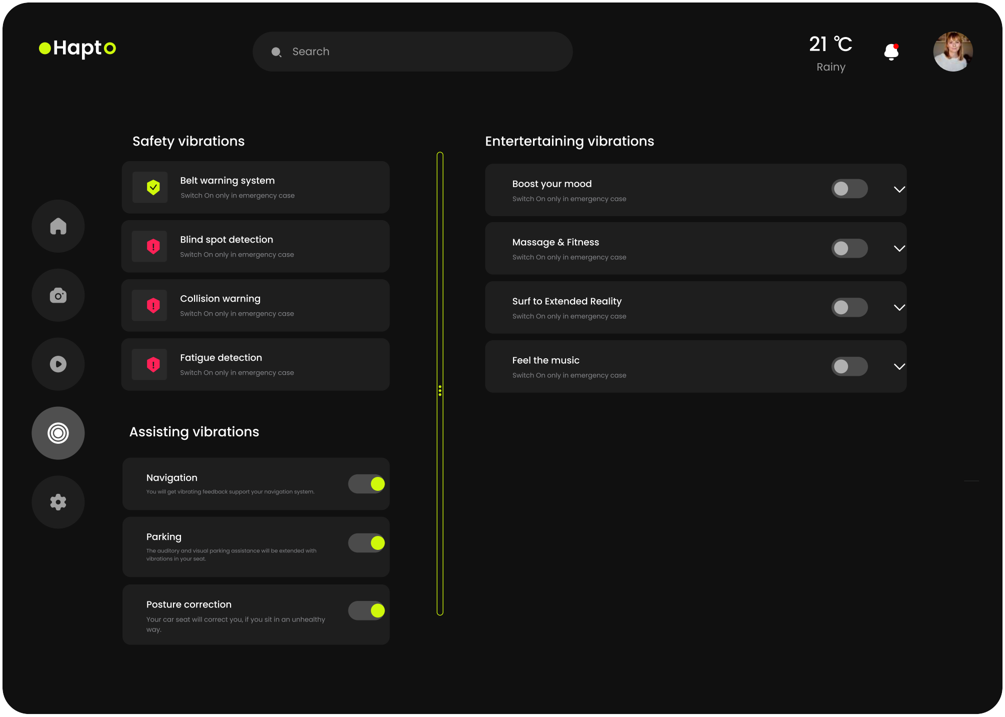
Choose a massage or a fitness mode

Select body part(s) to exercise

From the first to the second iteration (top image), massage vibration was repositioned from the Assisting Vibrations category to Entertaining Vibrations, alongside fitness vibration. In the third iteration (bottom image), **the amount of on-screen text was reduced to improve readability.**

Final Design ↘

This feature was **outside the project scope** and was developed independently as a self-initiated project.


Hapto
Search

21 ℃
Rainy

Safety vibrations

Belt warning system
Switch On only in emergency case

Blind spot detection
Switch On only in emergency case

Collision warning
Switch On only in emergency case

Fatigue detection
Switch On only in emergency case

Assisting vibrations

Navigation
You will get vibrating feedback support your navigation system.

Parking
The auditory and visual parking assistance will be extended with vibrations in your seat.

Posture correction
Your car seat will correct you, if you sit in an unhealthy way.

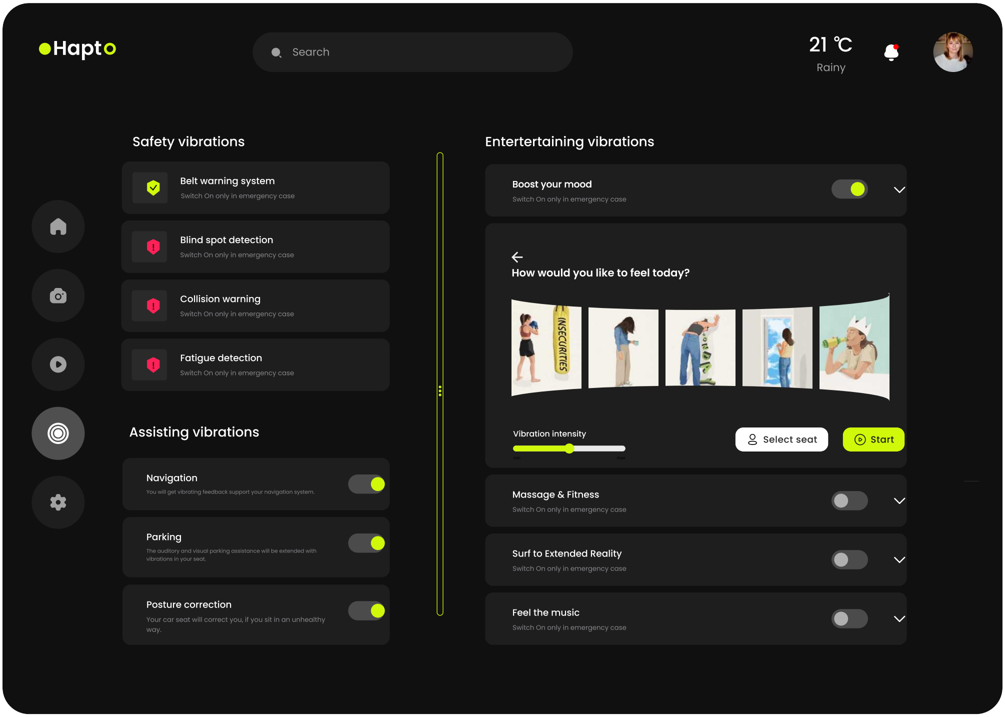
Entertertaining vibrations

Boost your mood
Switch On only in emergency case

Massage & Fitness
Switch On only in emergency case

Surf to Extended Reality
Switch On only in emergency case

Feel the music
Switch On only in emergency case


Hapto
Search
INSECURITIES
Select seat
Start

21 ℃
Rainy

Safety vibrations

Belt warning system
Switch On only in emergency case

Blind spot detection
Switch On only in emergency case

Collision warning
Switch On only in emergency case

Fatigue detection
Switch On only in emergency case

Assisting vibrations

Navigation
You will get vibrating feedback support your navigation system.

Parking
The auditory and visual parking assistance will be extended with vibrations in your seat.

Posture correction
Your car seat will correct you, if you sit in an unhealthy way.

Entertertaining vibrations

Boost your mood
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How would you like to feel today?

Vibration intensity

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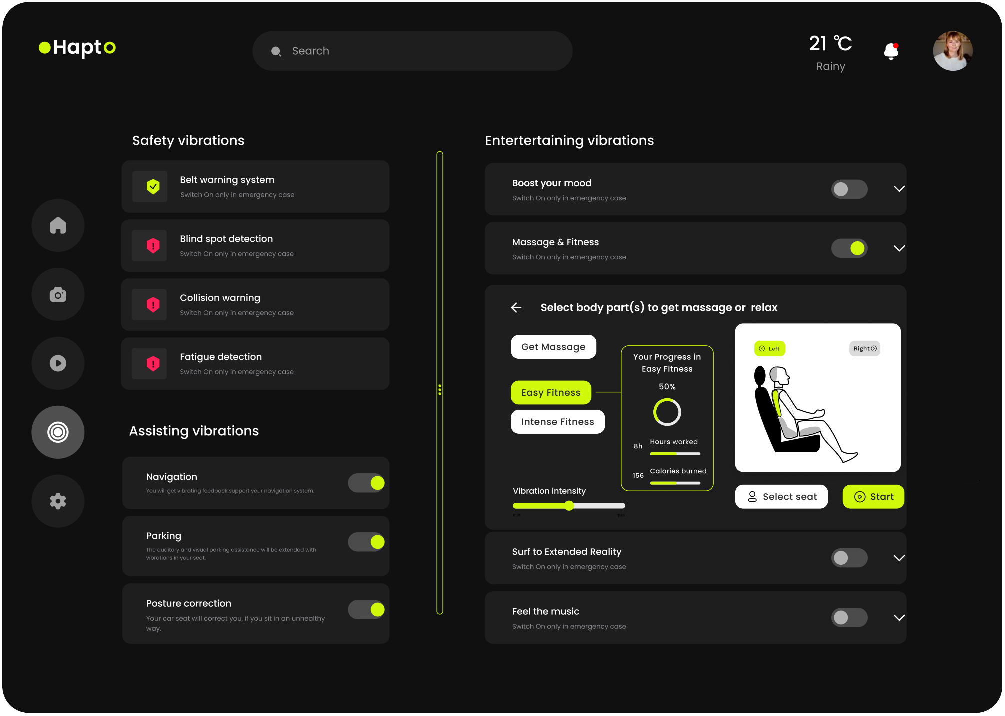

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Select body part(s) to get massage or relax

Get Massage

Easy Fitness

Intense Fitness

Your Progress in Easy Fitness
50%
8h Hours worked
156 Calories burned

Vibration intensity

Left
Right

Select seat

Start

Surf to Extended Reality
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Feel the music
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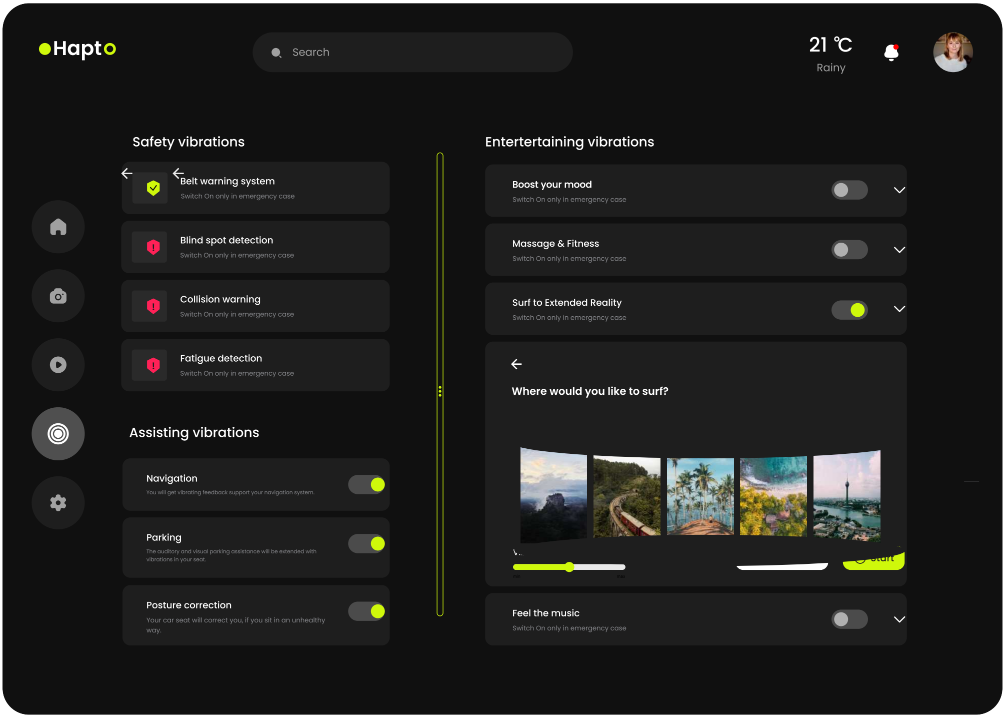

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Where would you like to surf?

Feel the music
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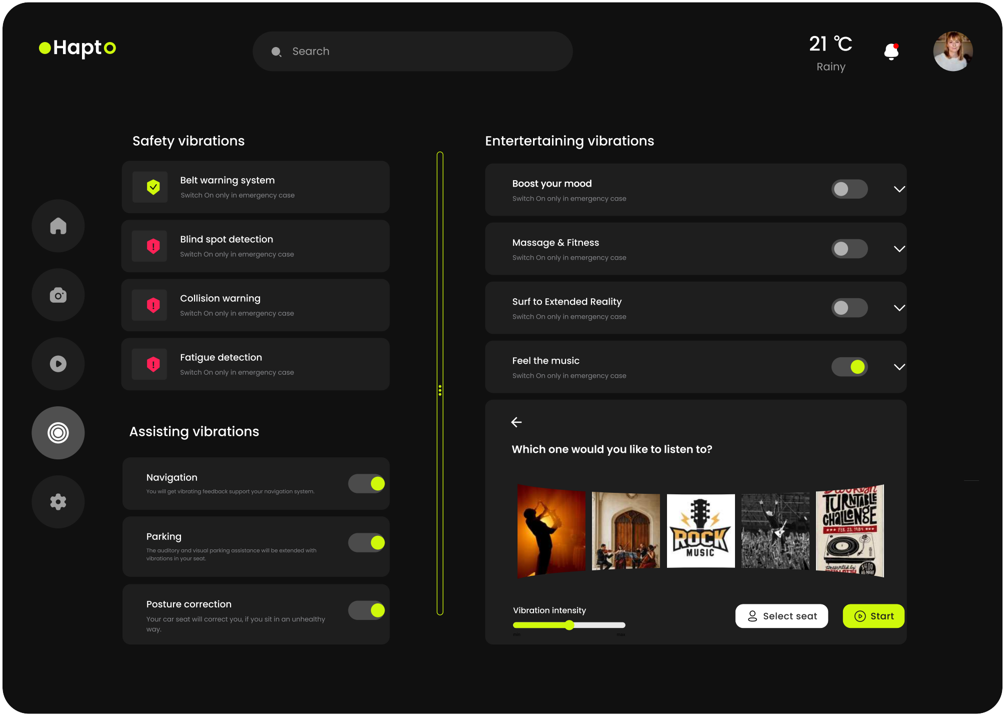

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Feel the music
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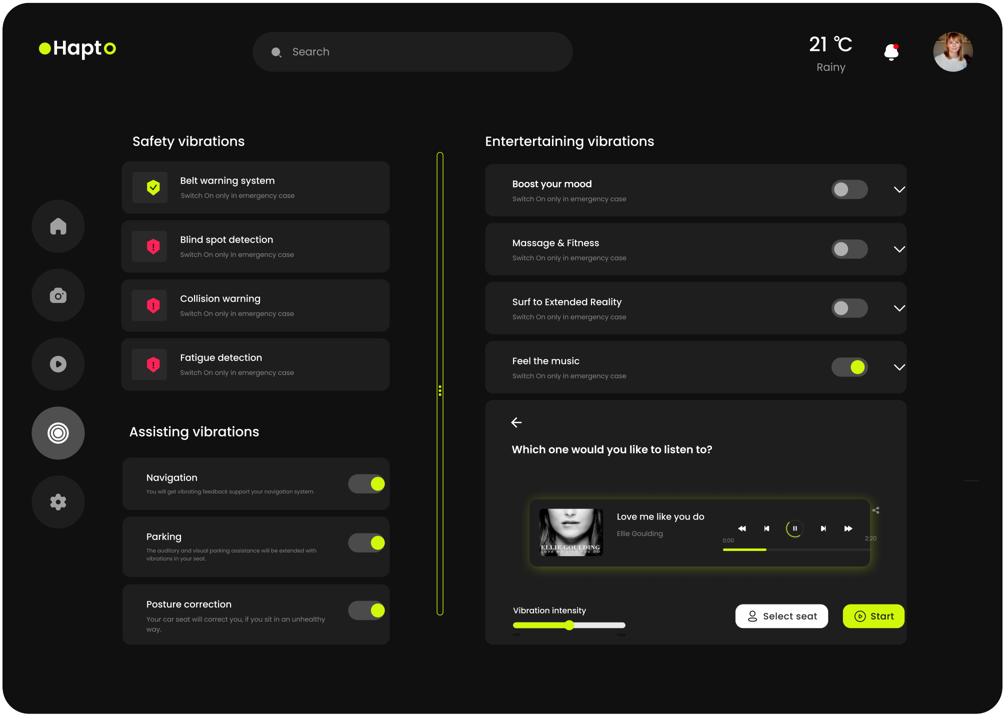
Which one would you like to listen to?

ROCK MUSIC
TURNTABLE CHALLENGE

Vibration intensity

Select seat
Start



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Which one would you like to listen to?

Love me like you do
Ellie Goulding
0:00
2:20

ELLIE GOULDING

Vibration intensity

Select seat
Start